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Evaluation of LABOR Synthesizer for Educational Purposes in Electronics

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Master of Science Thesis

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Preface

This thesis template is intended for honours, masters or PhD students at the Australian National University (ANU) who wish to write their thesis using the Quarto document format. It is highly recommended for students who code using Python, R or Julia and have many computational or analysis results in their thesis.

Note

This thesis template is available on GitHub at github.com/anuopensci/quarto-anu-thesis.

Benefits

The benefits of using Quarto document include:

- It allows you to write your thesis in a simple markup language called Markdown. This means that you can focus on writing your thesis without having to worry about formatting.
- The document can be output to a variety of formats including PDF, HTML, Word, and LaTeX.
- Code can be easily embedded in the document and executed. This means that you can include the results of your analysis in your thesis without having to manually copy and paste them. This is a good reproducible and scientific practice.
- You can easily integrate with aspects of GitHub (edit, reporting an issue, etc).

The above outlined benefits can also be considered as best practice. Version controlling and collaborative writing (via Git and GitHub) are important in managing multiple versions of your thesis and in collaborating with your supervisory team. Embedding code in your thesis is a good practice in reproducible research. Making your thesis in HTML format can allow for interactive web elements to be embedded while PDF format can be for general distribution and printing.

Getting started

There are several systems that you are expected to know to use this template. These include:

- Markdown syntax for writing
- Quarto or R Markdown syntax (note that these works for Python or Julia too) for embedding code
- (Optional) Git and GitHub for hosting

Frequently asked questions

What about Overleaf?

ANU has a professional account for Overleaf, which is great for those that use LaTeX regularly. Unfortunately, there is no equivalent system with track changes in Quarto. You can output the tex file from Quarto document and use this in Overleaf. The changes made in this tex document however has to be manually transferred back to the Quarto document. If your main output is mainly mathematical and you have little to no code outputs, Overleaf is probably a better choice.

Dedication

”Human nature is like water.

It takes the shape of its container.”

Wallace Stevens

Declaration of Authorship

https://www.hs-bremen.de/assets/hsb/de/Dokumente/ZLL/MMCC/declaration_indep_work_Nov_2024_Form.pdf

I hereby certify that I have written this thesis independently and have not used any sources or aids other than those specified. Sources and aids other than those indicated. The passages in the thesis that have been taken from other works in terms of wording or meaning are from other works are identified as such.

This statement also applies to the graphics, sketches and illustrations contained in the thesis. I have not already submitted the work in the same or a similar form, including extracts, as part of an examination or coursework.

I hereby certify that the electronic version of the dissertation submitted is a complete copy of the printed version. and agree to an electronic check of the thesis using plagiarism software.

Please tick the box: One of the two options must be chosen in consultation between the assessor and the assessee.

- ☐ Option 1: Use of AI-based tools without mandatory labelling I confirm that we have not used any AI-based tools whose use has not been agreed in writing with the verifier. I understand that the use of text or other content and products generated by AI-based tools is not a guarantee of their quality. I accept responsibility for the work submitted and the content contained therein. I also assure you that my creative influence predominates in this work.
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First name and surname:

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Abstract

Acknowledgements

I would like to express my sincere gratitude to my dog, Chuckles, for eating my research notes multiple times. If it wasn't for you, I would have finished this thesis earlier.

Chapter 1

Introduction

Reproducible research is an essential paradigm that promotes the idea that scientific investigations should be transparent, verifiable, and accessible to others. In an era where the scientific community faces concerns about the replicability of research findings, adopting reproducible research practices becomes imperative.

Chapter 2

Method

This report underscores the transformative impact of reproducible research on the scientific landscape, promoting a commitment to openness and accountability for the benefit of the entire research community.

References

Appendix A

Tools

This thesis was written using Quarto